



DATA PREPROCESSING



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Introduction

Real-world DBs are highly noisy, missing and inconsistent data.

Preprocessing techniques are used to improve the quality, efficiency and mining results

- **Data Cleaning:** remove noise and correct the inconsistencies in data
- **Data Integration:** merges data from multiple sources into coherent data store such as a data warehouse.
- **Data Reduction:** reduce data size by performing aggregations, eliminating redundant features and clustering.
- **Data Transformations:** where data are scaled to fall within a smaller range like 0.0 to 1.0, this can improve the accuracy and efficiency of mining algorithms involving distance measures.

Why Preprocess the Data?

- **Data Quality:** Data have quality if they satisfy the requirements of the intended use.
- Many factors comprising data quality
 - Accuracy
 - Completeness
 - Consistency
 - Timeliness
 - Believability
 - Interpretability

Cont.,

- Several attributes for various tuples have no record values, that is called **missing data** that reduces the quality by reporting errors, unusual values and inconsistencies
- The data you wish to analyze by the DM techniques are
 - Incomplete (lacking attribute values or containing aggregate data)
 - Inaccurate or noisy (having incorrect attr values that deviate from the expected)
 - Inconsistent (contains discrepancies in the dept codes used to categorize items)

Accuracy, Completeness and Consistency

- Inaccurate, incomplete and inconsistent data are common properties of large dbs and dws
- Reasons :
 - Data collection instruments used may be faulty
 - Human errors or computer errors occurring at data entry
 - Users may purposely submit incorrect data values for mandatory fields when they do not submit personal information(DoB)
 - Errors in data transmission
 - Technology limitations such as limited buffer size for coordinating synchronized data transfer and computation
 - Incorrect data may result in inconsistencies in naming conversions or formats in input fields(Date)

Timeliness

- **Timeliness:** also affects data quality
 - **All Electronics-** Update sales details at the month end
 - Some sales managers not update before month last day
 - And updated details have corrections and adjustments
 - Fact is month end data are not updated in a timely fashion has a negative impact on data quality

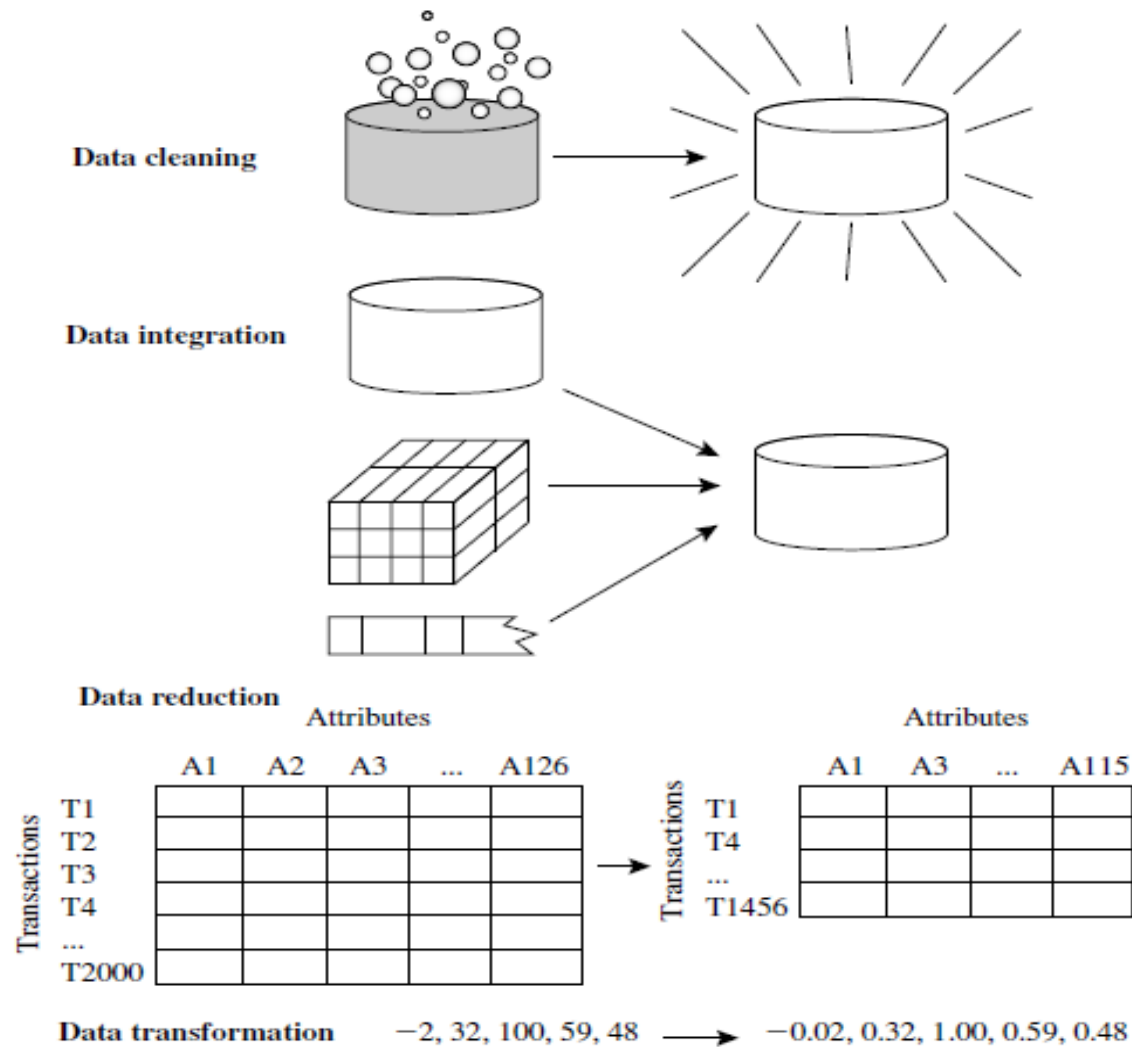
Believability and Interpretability

- Believability reflects how much the data are trusted by users/employees
- Interpretability reflects how easy the data are understood

Major Tasks in Data Preprocessing

- Data Cleaning
- Data Integration
- Data Reduction
- Data Transformation

Data Preprocessing



Data Cleaning

- Data cleaning routines attempt to
 - Fill in missing values
 - smooth out noise while identifying outliers
 - correct inconsistencies in the data
 - Resolve redundancy caused by data integration

Missing Values

1. Ignore the tuple,
2. Fill in the missing value manually- time consuming may not be feasible for large datasets
3. Use a global constant to fill in the missing value- like *unknown* or ∞
4. Use a measure of central tendency- mean for symmetric data and median for skewed data
5. Use the attribute mean or median for all samples belonging to the same class as the given tuple
6. Use the most probable value to fill in the missing value
 - Determined with regression, inference based tools using a Bayesian formalism or decision tree induction

Noisy Data

- Noise is a random error or variance in a measured variable
 - Boxplots and scatter plots are used to identifying the outliers which may represent noise
 - Ex: attribute “*price*” , we have to smooth out the data to remove noise

Smoothing techniques

- **Binning:** smooth the sorted data value by consulting its neighbourhood i.e. the values around it
- The sorted values are distributed into a no.of buckets or bins and the perform local smoothing

- Smooth by Bin mean
- Smooth by bin medians
- Smooth by bin boundaries(min and max values are as bin boundaries)

Sorted data for *price* (in dollars): 4, 8, 15, 21, 21, 24, 25, 28, 34

Partition into (equal-frequency) bins:

Bin 1: 4, 8, 15

Bin 2: 21, 21, 24

Bin 3: 25, 28, 34

Smoothing by bin means:

Bin 1: 9, 9, 9

Bin 2: 22, 22, 22

Bin 3: 29, 29, 29

Smoothing by bin boundaries:

Bin 1: 4, 4, 15

Bin 2: 21, 21, 24

Bin 3: 25, 25, 34

- **Regression:** is a technique that conforms data values to a function
- **Linear Regression** involves finding the **best** line to fit two attributes so that one attribute can be used to predict the other
- **Multiple linear regression** is an extension of linear regression, where more than two attributes are involved and data are fit to multidimensional surface

- **Outlier analysis:** detected by clustering
- **Ex:** similar values are organized into groups or clusters, values outside of the set of clusters are considered as

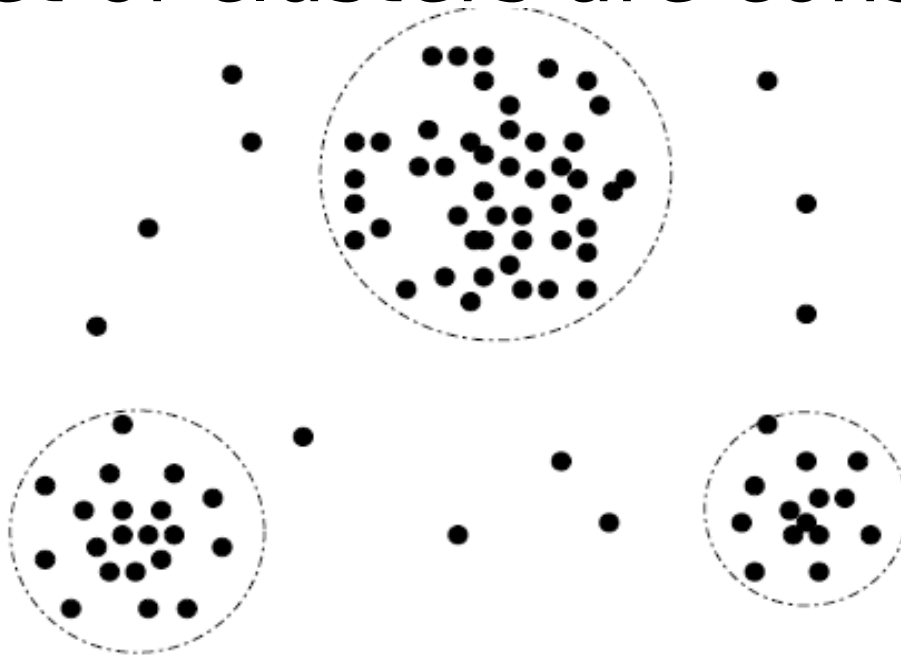


Figure 3.3 A 2-D customer data plot with respect to customer locations in a city, showing three data clusters. Outliers may be detected as values that fall outside of the cluster sets.

Data cleaning as a Process

- First step is **discrepancy detection** (similarity or difference)
- **Discrepancy** caused several factors
 - Poorly designed data entry forms having many optional fields
 - Human errors in data entry
 - Deliberate errors(not want to give)
 - Data decay(Out dated address)
 - Errors in instrumentation devices that record data
 - Data integration process may cause

- **Data discrepancy detection**
 - **Use Metadata** (knowledge about data, data type and domain of attribute, acceptable values)
 - Taking care about inconsistent data representations (“2020/01/03” and “03/01/2020”)
 - **Field overloading** – is an error source it results when developers squeeze new attribute definition into unused portions of already defined attributes (used 31 bits out of 32 bits)

- Data is examined for **unique rules, constructive rules and null rules**
 - Unique rule: Each value of the given attribute must be different
 - Constructive rule: There can be no missing values between the lowest and highest values and all values must be unique also
 - Null rule: specifies the use of blanks, question marks, special characters or other strings that may indicate the null condition

- Once we find data discrepancies, we need to define and apply series of transformations to correct them
- Commercial tools assist in the data transformation step
- **Data Migration tools:** allow simple transformations to be specified (replace gender by sex)
- **ETL (Extraction/transformation/loading) tools:** allow users to specify transforms through a GUI

Data Integration

- The semantic **heterogeneity and structure of the data** pose great challenges in the data integration
 - Entity identification problem – How can we match schema and objects from different sources?
 - Correlation tests on numeric data and nominal data – Specifies the correlation between objects

Entity Identification Problem

- No. of issues are considered during data integration
 - Schema integration and object matching can be tricky
- The equivalent of real world entities from multiple data sources is known as **entity identity problem**
 - Ex: Different representations and Different scales like cust_id and customer_id how they are referred
 - Metadata for each attribute (name, meaning, datatype, range of values permitted for the attribute, null rules for handling blanks, zeros)
 - Such metadata may also be used to help transform the data.

Redundancy and Correlation Analysis

- An attribute may be redundant if it can be derived from another attribute.
- Careful integration of data from multiple sources may help/avoid redundancies
- Some redundancies can be detected by correlation analysis
- Given two attributes, correlation analysis can measure how strongly one attribute implies the other, based on the available data
 - For Nominal data, we use Chi-Square test
 - For Numeric attribute, we use Correlation Coefficient and Covariance

χ^2 Correlation Test for Nominal data

- Correlation relationship between two attributes, A and B, can be discovered by Chi-Square test
- Suppose A has c distinct values, namely $a_1, a_2, \dots, a_c$
- B has r distinct values, namely $b_1, b_2, \dots, b_r$
- The data tuples described by contingency table, with the **c** values of A making up the **columns**, **r** values of B making up the **rows**.

$$\text{Chi-Square} = \sum_{i=1}^c \sum_{j=1}^r \frac{(O_{ij} - e_{ij})^2}{e_{ij}}$$

- O_{ij} is Observed frequency(actual count) of the joint event (A_i, B_j)
- e_{ij} is Expected frequency

$$e_{ij} = \frac{\text{count}(A=a_i) \times \text{count}(B=b_j)}{n}$$

- n is the total number of data tuples

χ^2 Correlation Test for Nominal data

- Chi-Square statistic tests the hypothesis that A and B are independent, i.e., there is no correlation b/w them
- The test is based on the significant level, with $(r-1) \times (c-1)$ degree of freedom.
- If the hypothesis can be rejected, then we say that A and B are statistically correlated.
- In terms of a p-value and a chosen significance level (α), the test can be interpreted as follows:
 - **If p-value $\leq \alpha$:** significant result, reject null hypothesis (H_0), dependent.
 - **If p-value $> \alpha$:** not significant result, fail to reject null hypothesis (H_0), independent.

Example

	male	female	Total
<i>fiction</i>	250 (90)	200 (360)	450
<i>non_fiction</i>	50 (210)	1000 (840)	1050
Total	300	1200	1500

Note: Are gender and preferred_reading correlated?

- The observed frequency (or count) of each possible joint event is summarized in the contingency table, where the numbers in parentheses are the expected frequencies.
- $e_{11} = \frac{\text{count}(\text{male}) \times \text{count}(\text{fiction})}{n} = \frac{450 \times 300}{1500} = 90$
- Compute Chi-Square value:

$$\begin{aligned}\chi^2 &= \frac{(250 - 90)^2}{90} + \frac{(50 - 210)^2}{210} + \frac{(200 - 360)^2}{360} + \frac{(1000 - 840)^2}{840} \\ &= 284.44 + 121.90 + 71.11 + 30.48 = 507.93.\end{aligned}$$

Correlation Coefficient for Numeric Data

Pearson's product moment coefficient

$$r_{A,B} = \frac{\sum_{i=1}^n (a_i - \bar{A})(b_i - \bar{B})}{n\sigma_A\sigma_B} = \frac{\sum_{i=1}^n (a_i b_i) - n\bar{A}\bar{B}}{n\sigma_A\sigma_B},$$

- n is the number of tuples
- a_i and b_i are respective values of A and B in tuple i
- $\bar{A}$ and $\bar{B}$ are respective mean values of A and B
- σ_A and σ_B are respective standard deviation of A and B
- $\sum(a_i, b_i)$ is the sum of the AB cross-product
- **Note:** $-1 \leq r_{A,B} \leq +1$
- If $r_{A,B}$ is greater than 0, then A and B are Positively Correlated (values of A increase as the values of B increase)
- If the resulting value = 0, then A and B are independent, and there is no correlation between them
- If the resulting value is < 0 , then A and B are Negatively correlated (values of A increase as the values of B decrease)

Covariance of Numeric Data

- In P&S, correlation and covariance are similar measures for assessing how much two attributes change together
- Consider two numeric attr A and B, and a set of n observations.
- The mean values of A and B are also known as **Expected values** of A and B, i.e., $E(A)$ and $E(B)$ respectively.
- The covariance between A and B is defined as

$$Cov(A, B) = E((A - \bar{A})(B - \bar{B})) = \frac{\sum_{i=1}^n (a_i - \bar{A})(b_i - \bar{B})}{n}.$$

- If we compare correlation coefficient with covariance, we can see that $r_{A,B} = \frac{Cov(A,B)}{\sigma_A \sigma_B}$
- Covariance can also be shown as $Cov(A, B) = E(A \cdot B) - \bar{A}\bar{B}.$

Covariance of Numeric Data

- If **$COV(A,B)$** is greater than 0, then A and B are Positively Correlated (values of A increase as the values of B increase)
- If the resulting value = 0, then A and B are independent, and there is no correlation between them
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Example

Stock Prices for *AllElectronics* and *HighTech*

<i>Time point</i>	<i>AllElectronics</i>	<i>HighTech</i>
t1	6	20
t2	5	10
t3	4	14
t4	3	5
t5	2	5

- If the stocks are affected by the same industry trends, will their prices rise or fall together?

Tuple Duplication

- The use of denormalized tables (often done to improve performance by avoiding joins) is another source of data redundancy
- Inconsistencies often arise between various duplicates, due to inaccurate data entry or updating some but not all data occurrences.

DATA TRANSFORMATION

- **Smoothing:** To remove noise from the data (Techniques include binning, regression and clustering)
- **Attribute construction (or *feature construction*):** New attributes are constructed from the given set of attributes to help the mining process.
- **Aggregation:** Aggregation operations are applied to the data. Ex- the daily sales data may be aggregated so as to compute monthly and annual total amounts.
- **Normalization:** The attribute data are scaled so as to fall within a smaller range, such as -1.0 to 1.0, or 0.0 to 1.0.
- **Discretization:** The raw values of a numeric attribute (e.g., *age*) are replaced by interval labels (e.g., 0-10, 11-20, etc.) or conceptual labels (e.g., *youth*, *adult*, *senior*). The labels, in turn, can be recursively organized into higher-level concepts, resulting in a *concept hierarchy* for the numeric attribute.
- **Concept hierarchy generation for nominal data:** Attributes such as *street* can be generalized to higher-level concepts, like *city* or *country*. Many hierarchies for nominal attributes are implicit within the database schema and can be automatically defined at

Data Transformation by Normalization

- Changing the measurement unit of attribute values/units is called normalization
- Transforming the data to fall within a smaller range such as $[-1,1]$ or $[0.0,1.0]$
- Normalization is particularly useful for classification algorithms involving neural networks or distance measurements such as clustering
- Data Normalization methods
 - **Min-max normalization**
 - **Z-score normalization**
 - **Normalization by Decimal Scaling**

Min-Max Normalization

- Performs a linear transformation on the original data.
- Suppose that $\min_A$ and $\max_A$ are the minimum and maximum values of an attribute, A .
- Min-max normalization maps a value, v_i , of A to v'_i in the range $[new_min_A, new_max_A]$ by computing

$$v' = \frac{v - \min_A}{\max_A - \min_A} (new_max_A - new_min_A) + new_min_A$$

- Min-max normalization preserves the relationships among the original data values.
- Ex. Let income range \$12,000 to \$98,000 normalized to [0.0, 1.0]. Then \$73,000 is mapped to

$$\frac{73,600 - 12,000}{98,000 - 12,000} (1.0 - 0) + 0 = 0.716$$

Z(Zero) - Score Normalization

- The values for an attribute A , are normalized based on mean and standard deviation of A .
- A value, v_i , of A *normalized* to v'_i by computing
$$v' = \frac{v - \mu_A}{\sigma_A}$$
- Ex: the mean and standard deviation of the values for the attribute income are \$54,000 and \$16,000.
- Find the normalized value of income \$73,600?

Normalization by Decimal Scaling

- Normalizes by moving the decimal point of values of attribute A
- The number of decimal points moved depends on the maximum absolute value of A.
- A value, v_i , of A *normalized* to v'_i by computing

$$v' = \frac{v}{10^j}$$

- Where j is the smallest integer, $\text{Max}(|v'|) < 1$

Data Reduction

- To obtain a reduced representation of the data set that is much smaller in volume but yet produce the same (almost same) analytical results.
- Why data reduction? — A database/data warehouse may store terabytes of data. Complex data analysis and mining may take a very long time to run on the complete data set.
- Data reduction strategies
 - Dimensionality reduction, e.g., reduces the no.of random variables or attributes
 - Wavelet transforms
 - Principal Components Analysis (PCA)
 - Feature subset selection, feature creation
 - Numerosity reduction, Replace the original data volume by alternative or smaller forms
 - Parametric methods: Regression and Log-Linear Models
 - Non - parametric methods: Histograms, clustering, sampling
 - Data compression:
 - Lossy

Wavelet Transforms

- The **discrete wavelet transform (DWT)** is a linear signal processing technique that, when applied to a data vector $\mathbf{X}$, transforms it to a numerically different vector, $\mathbf{X}'$, of **wavelet coefficients**. The two vectors are of the same length. When applying this technique to data reduction, we consider each tuple as an n -dimensional data vector, that is, $\mathbf{X} = (x_1, x_2, \dots, x_n)$, depicting n measurements made on the tuple from n database attributes

- ***“How can this technique be useful for data reduction if the wavelet transformed data are of the same length as the original data?”***
- The usefulness lies in the fact that the wavelet transformed data can be truncated.
- A compressed approximation of the data can be retained by storing only a small fraction of the strongest of the wavelet coefficients.
- Ex: all wavelet coefficients larger than some user-specified threshold can be retained. All other coefficients are set to 0.
 - The resulting data representation is therefore very sparse, so that operations that can take advantage of data sparsity are computationally very fast if performed in wavelet space.
- The technique also works to remove noise without smoothing out the main features of the data, making it effective for data cleaning as well.
- Given a set of coefficients, an approximation of the original data can be constructed by applying the *inverse* of the DWT used.

- DWT achieves better lossy compression.
- There are several families of DWT
 - Haar
 - Daubechies-4
 - Daubechies-6
- DWT uses a **hierarchical pyramid algorithm**, that halves the data at each iteration, resulting in a fast computational speed

Hierarchical Pyramidal Algorithm

- The length, L , of the input data vector must be an integer power of 2. This condition can be met by padding the data vector with zeros as necessary ($L \geq n$).
- Each transform involves applying two functions. The first applies some data smoothing, such as a sum or weighted average. The second performs a weighted difference, which acts to bring out the detailed features of the data.
- The two functions are applied to pairs of data points in $\mathbf{X}$, that is, to all pairs of measurements (x_{2i}, x_{2i+1}) . This results in two data sets of length $L/2$. In general, these represent a smoothed or low-frequency version of the input data and the high frequency content of it, respectively.
- The two functions are recursively applied to the data sets obtained in the previous loop, until the resulting data sets obtained are of length 2.
- Selected values from the data sets obtained in the previous iterations are designated the wavelet coefficients of the transformed data.

Principle Component Analysis (Karhunen-Loeve, or K-L, method)

- Suppose that the data to be reduced consist of tuples or data vectors described by n attributes or dimensions.
- **Principal components analysis** searches for k n -dimensional orthogonal vectors that can best be used to represent the data, where $k \leq n$.
- The original data are thus projected onto a much smaller space, resulting in dimensionality reduction.
- Unlike attribute subset selection which reduces the attribute set size by retaining a subset of the initial set of attributes, PCA “combines” the essence of attributes by creating an alternative, smaller set of variables.
- The initial data can then be projected onto this smaller set. PCA often reveals relationships that were not previously suspected and thereby allows interpretations that would not ordinarily result.

PCA Procedure

- The input data are normalized, so that each attribute falls within the same range. This step helps ensure that attributes with large domains will not dominate attributes with smaller domains.
- PCA computes k orthonormal vectors that provide a basis for the normalized input data. These are unit vectors that each point in a direction perpendicular to the others. These vectors are referred to as the *principal components*. The input data are a linear combination of the principal components.
- The principal components are sorted in order of decreasing “significance” or strength. The principal components essentially serve as a new set of axes for the data, providing important information about variance. That is, the sorted axes are such that the first axis shows the most variance among the data, the second axis shows the next highest variance, and so on.
- Because the components are sorted in decreasing order of “significance,” the data size can be reduced by eliminating the weaker components, that is, those with low variance. Using the strongest principal components, it should be possible to reconstruct a good approximation of the original data.

PCA - Advantages

- PCA can be applied to ordered and unordered attributes, and can handle sparse data and skewed data.
- Multidimensional data of more than two dimensions can be handled by reducing the problem to two dimensions.
- Principal components may be used as inputs to multiple regression and cluster analysis.
- In comparison with wavelet transforms, PCA tends to be better at handling sparse data, whereas wavelet transforms are more suitable for data of high dimensionality.

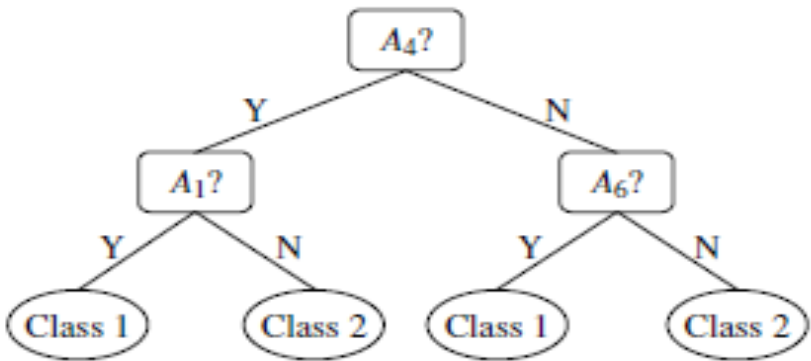
Attribute Subset Selection

- **Attribute subset selection** reduces the data set size by removing irrelevant or redundant attributes (or dimensions).
- The goal of attribute subset selection is to find a minimum set of attributes such that the resulting probability distribution of the data classes is as close as possible to the original distribution obtained using all attributes.
- It reduces the number of attributes appearing in the discovered patterns, helping to make the patterns easier to understand.

- “How can we find a ‘good’ subset of the original attributes?”
 - For n attributes, there are 2^n possible subsets. An exhaustive search for the optimal subset of attributes can be prohibitively expensive, especially as n and the number of data classes increase. Therefore, heuristic methods that explore a reduced search space are commonly used for attribute subset selection.
- These methods are typically **greedy** in that, while searching through attribute space, they always make what looks to be the best choice at the time.
- Their strategy is to make a locally optimal choice in the hope that this will lead to a globally optimal solution.
- Such greedy methods are effective in practice and may come close to estimating an optimal solution.
- The “best” (and “worst”) attributes are typically determined using tests of statistical significance, which assume that the attributes are independent of one another.
- Many other attribute evaluation measures can be used such as the *information gain* measure used in building decision trees for classification.

Greedy(Heuristic) methods for Attribute Subset Selection

- **Stepwise forward selection:**
 - starts with an empty set of attributes as the reduced set.
 - Best of the original attributes is determined and added to the reduced set.
 - At each subsequent iteration or step, the best of the remaining original attributes is added to the set.
- **Stepwise backward elimination:**
 - Starts with the full set of attributes.
 - At each step, it removes the worst attribute remaining in the set.
- **Combination of forward selection and backward elimination:**
 - The stepwise forward selection and backward elimination methods can be combined
 - At each step, the procedure selects the best attribute and removes the worst from among the remaining attributes.
- **Decision tree induction:**
 - Decision tree algorithms (e.g., ID3, C4.5, and CART) were originally intended for classification.
 - Decision tree induction constructs a flow chart like structure where each internal (nonleaf) node denotes a test on an attribute, each branch corresponds to an outcome of the test, and each external (leaf) node denotes a class prediction.
 - At each node, the algorithm chooses the “best” attribute to partition the data into individual classes.
- When decision tree induction is used for attribute subset selection, a tree is constructed from the given data. All attributes that do not appear in the tree are assumed to be irrelevant. The set of attributes appearing in the tree form the reduced subset of attributes.

Forward selection	Backward elimination	Decision tree induction
Initial attribute set: $\{A_1, A_2, A_3, A_4, A_5, A_6\}$ Initial reduced set: $\{\}$ $\Rightarrow \{A_1\}$ $\Rightarrow \{A_1, A_4\}$ $\Rightarrow$ Reduced attribute set: $\{A_1, A_4, A_6\}$	Initial attribute set: $\{A_1, A_2, A_3, A_4, A_5, A_6\}$ $\Rightarrow \{A_1, A_3, A_4, A_5, A_6\}$ $\Rightarrow \{A_1, A_4, A_5, A_6\}$ $\Rightarrow$ Reduced attribute set: $\{A_1, A_4, A_6\}$	Initial attribute set: $\{A_1, A_2, A_3, A_4, A_5, A_6\}$  <pre> graph TD A4["A4?"] -- Y --> A1["A1?"] A4 -- N --> A6["A6?"] A1 -- Y --> C1_1((Class 1)) A1 -- N --> C2_1((Class 2)) A6 -- Y --> C1_2((Class 1)) A6 -- N --> C2_2((Class 2)) </pre> $\Rightarrow$ Reduced attribute set: $\{A_1, A_4, A_6\}$

Greedy (heuristic) methods for attribute subset selection.

Numerosity Reduction

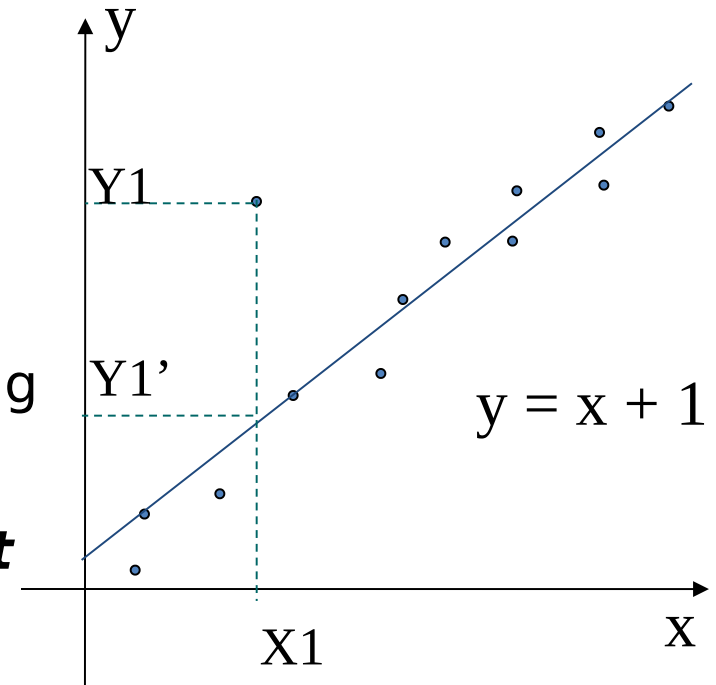
- Reduce data volume by choosing alternative, *smaller forms* of data representation
- **Parametric methods** (e.g., regression)
 - Assume the data fits some model, estimate model parameters, store only the parameters, and discard the data (except possible outliers)
 - Ex.: Log-linear models—obtain value at a point in m -D space as the product on appropriate marginal subspaces
- **Non-parametric** methods
 - Do not assume models
 - Major families: histograms, clustering, sampling, ...

Parametric Data Reduction: Regression and Log-Linear Models

- **Linear regression**
 - Data modeled to fit a straight line
 - Often uses the least-square method to fit the line
- **Multiple regression**
 - Allows a response variable Y to be modeled as a linear function of multidimensional feature vector
- **Log-linear model**
 - Approximates discrete multidimensional probability distributions

Regression Analysis

- Regression analysis: A collective name for techniques for the modeling and analysis of numerical data consisting of values of a **dependent variable** (also called **response variable** or *measurement*) and of one or more *independent variables* (aka. **explanatory variables** or **predictors**)
- The parameters are estimated so as to give a "**best fit**" of the data
- Most commonly the best fit is evaluated by using the **least squares method**, but other criteria



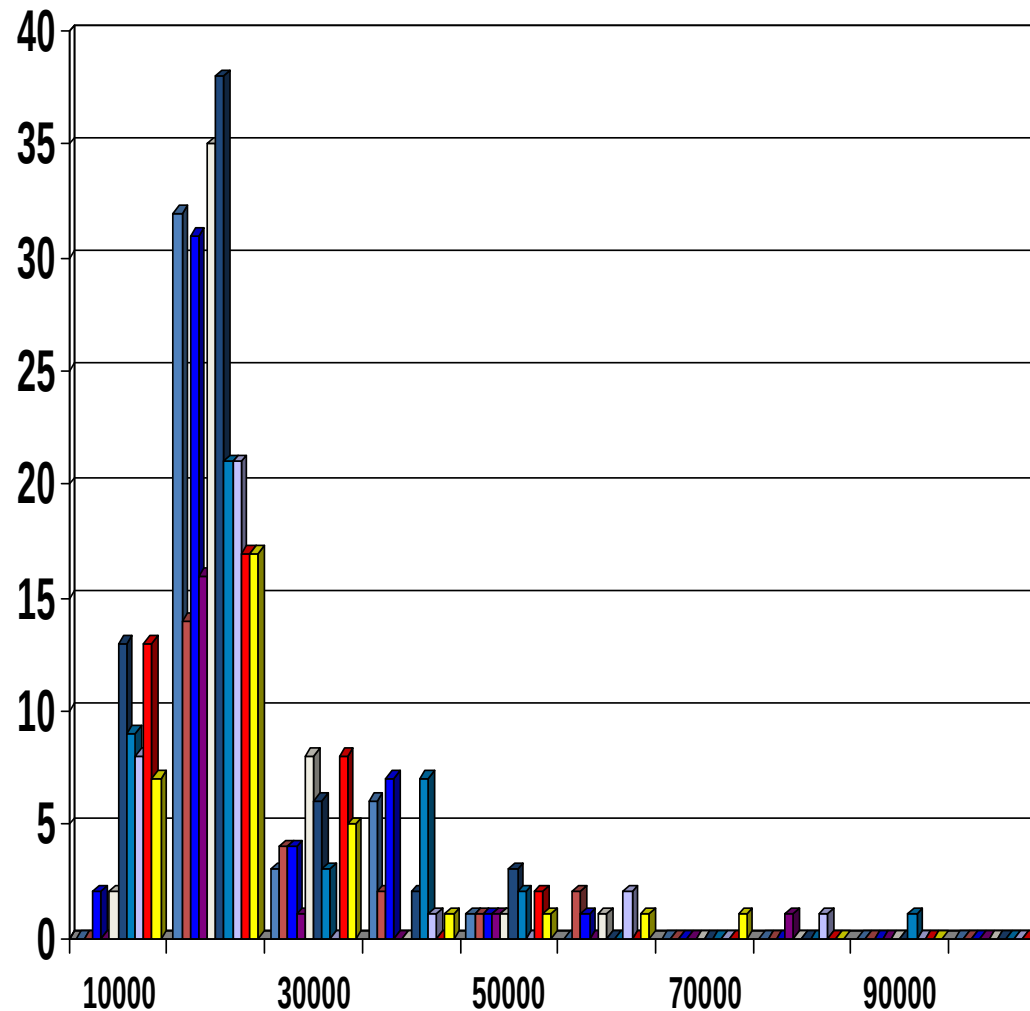
- Used for prediction (including forecasting of time-series data), inference, hypothesis testing, and modeling of causal relationships

Regress Analysis and Log-Linear Models

- Linear regression: $Y = w X + b$
 - Two regression coefficients, w and b , specify the line and are to be estimated by using the data at hand
 - Using the least squares criterion to the known values of $Y_1, Y_2, \dots, X_1, X_2, \dots$
- Multiple regression: $Y = b_0 + b_1 X_1 + b_2 X_2$
 - Many nonlinear functions can be transformed into the above
- Log-linear models:
 - Approximate discrete multidimensional probability distributions
 - Estimate the probability of each point (tuple) in a multi-dimensional space for a set of discretized attributes, based on a smaller subset of dimensional combinations
 - Useful for dimensionality reduction and data smoothing

Histogram Analysis

- Divide data into buckets and store average (sum) for each bucket
- Partitioning rules:
 - Equal-width: equal bucket range
 - Equal-frequency (or equal-depth)



Clustering

- Partition data set into clusters based on similarity, and store cluster representation (e.g., centroid and diameter) only
- Can be very effective if data is clustered but not if data is “smeared”
- Can have hierarchical clustering and be stored in multi-dimensional index tree structures
- There are many choices of clustering definitions and clustering algorithms
- Cluster analysis will be studied in depth in Chapter 10

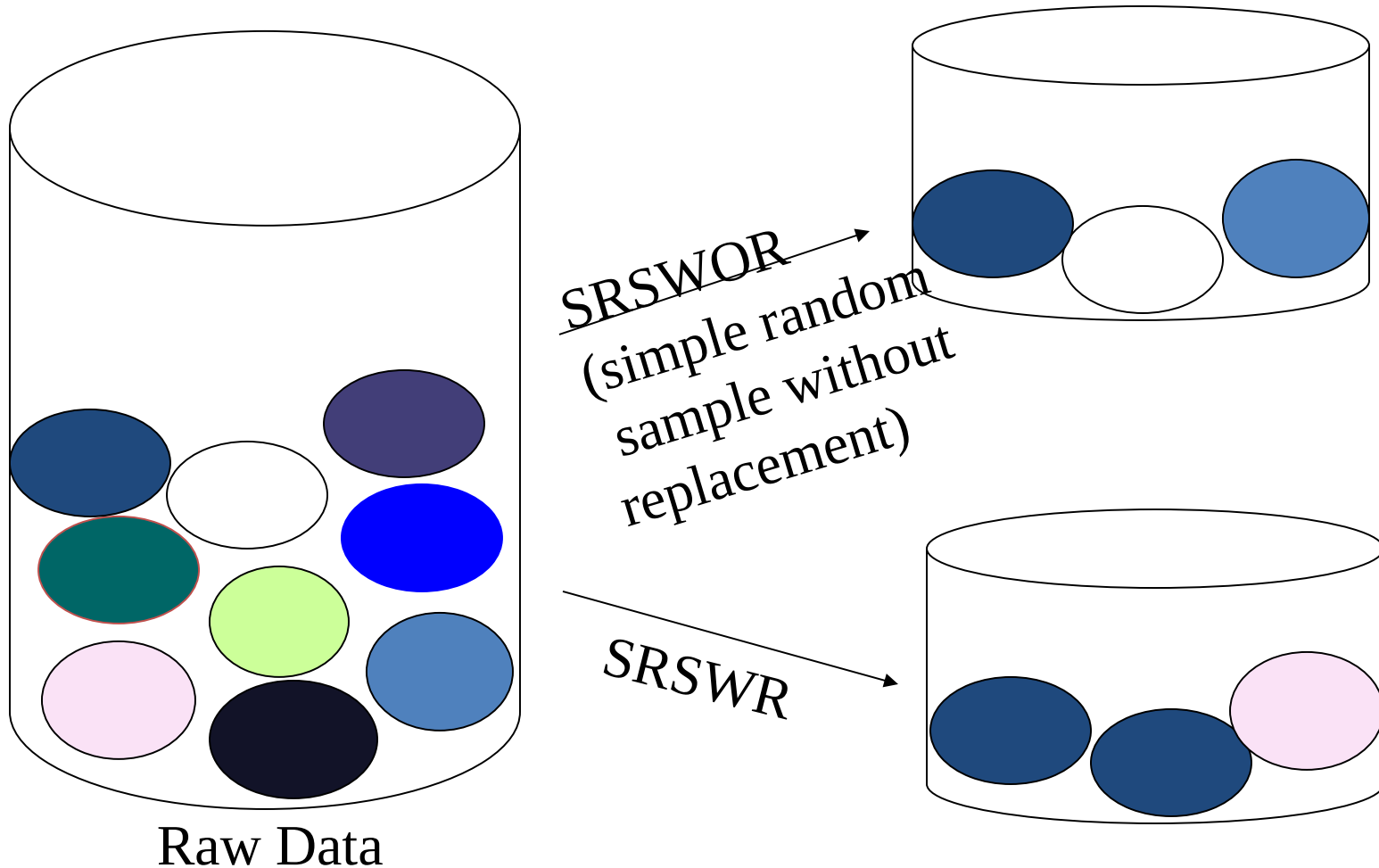
Sampling

- Sampling: obtaining a small sample s to represent the whole data set N
- Allow a mining algorithm to run in complexity that is potentially sub-linear to the size of the data
- Key principle: Choose a **representative** subset of the data
 - Simple random sampling may have very poor performance in the presence of skew
 - Develop adaptive sampling methods, e.g., stratified sampling:
- Note: Sampling may not reduce database I/Os (page at a time)

Types of Sampling

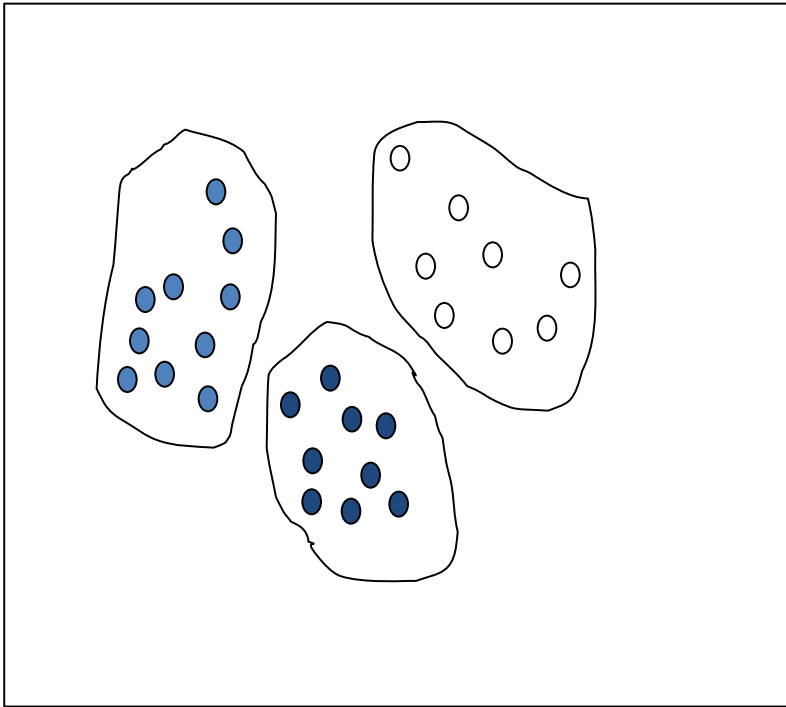
- **Simple random sampling**
 - There is an equal probability of selecting any particular item
- **Sampling without replacement**
 - Once an object is selected, it is removed from the population
- **Sampling with replacement**
 - A selected object is not removed from the population
- **Stratified sampling:**
 - Partition the data set, and draw samples from each partition (proportionally, i.e., approximately the same percentage of the data)
 - Used in conjunction with skewed data

Sampling: With or without Replacement

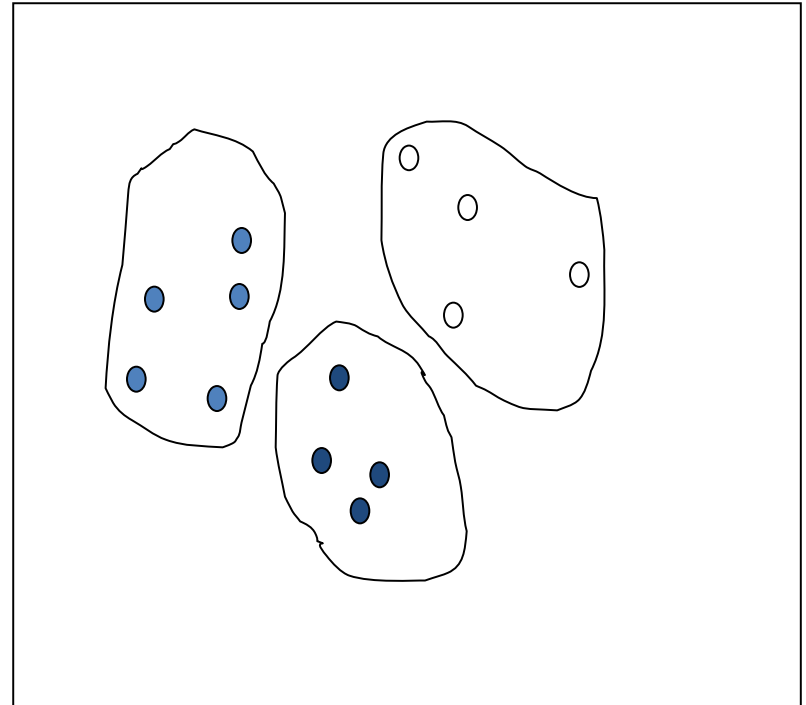


Sampling: Cluster or Stratified Sampling

Raw Data



Cluster/Stratified Sample



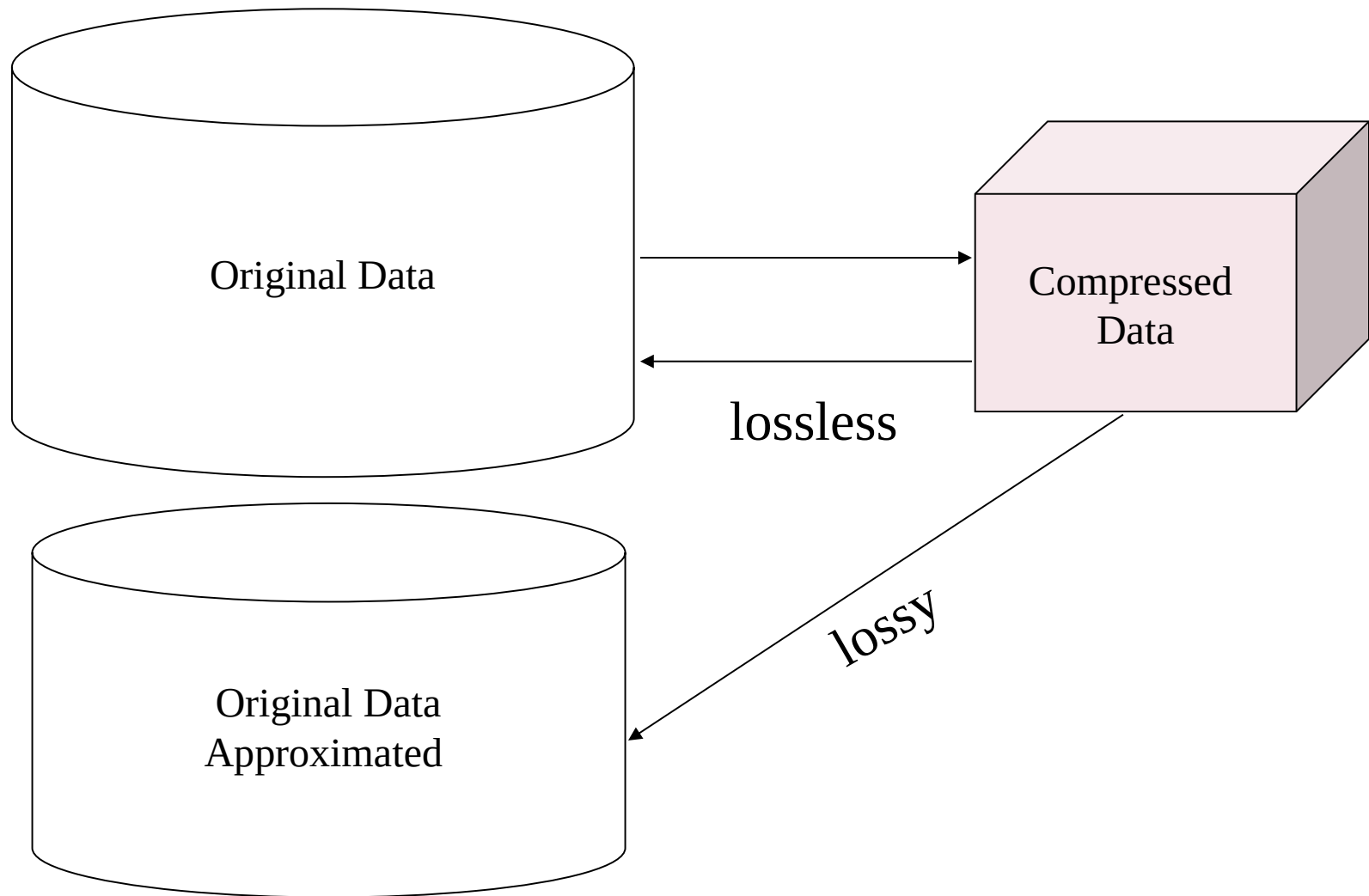
Data Cube Aggregation

- The lowest level of a data cube (base cuboid)
 - The aggregated data for an **individual entity of interest**
 - E.g., a customer in a phone calling data warehouse
- Multiple levels of aggregation in data cubes
 - Further reduce the size of data to deal with
- Reference appropriate levels
 - Use the smallest representation which is enough to solve the task
- Queries regarding aggregated information should

Data Compression

- String compression
 - There are extensive theories and well-tuned algorithms
 - Typically lossless, but only limited manipulation is possible without expansion
- Audio/video compression
 - Typically lossy compression, with progressive refinement
 - Sometimes small fragments of signal can be reconstructed without reconstructing the whole
- Time sequence is not audio
 - Typically short and vary slowly with time
- Dimensionality and numerosity reduction may also be considered as forms of data compression

Data Compression



Thank You